Temperature, enzymes and metabolism in *E. coli*

What this model adds to what is published

Living Systems Institute, University of Exeter

2026-09-11

The question

- Metabolism is temperature-dependent; **thermal performance curves are the phenotype**
- An **etcGEM** predicts a whole TPC from *per-enzyme* thermodynamics — no curve fitting to the shape
- Three reference points, and every claim here says which one it is relative to:
  - **Li et al. 2021**[1] — the only published etcGEM (yeast)
  - **Pettersen & Almaas 2023**[2] — re-ran it and found the inference unstable
  - **Madkaikar 2023**[3] — the MRes baseline this group inherited
- **What is new here: the data that makes the inference identifiable, and the audits that say what a fitted etcGEM does and does not know**

What an etcGEM is

- A genome-scale metabolic model of ~2 500 reactions, plus three things:
  - **an enzyme budget** — carrying flux costs protein, and protein is finite
  - **per-enzyme** $k_{cat}(T)$ — each enzyme's turnover peaks at its own T_{opt} and falls away either side (macromolecular rate theory, curvature $\Delta C_p^{\ddagger}$)[4]
  - **per-enzyme unfolding** — the folded fraction collapses at each enzyme's melting temperature T_m
- Warm the cell and every enzyme moves on its own curve; the LP re-allocates the budget
- **The organism's TPC is then emergent** — it is an output, not a fitted function

The two temperature responses, per enzyme

Left: turnover $\hat{k}(T)$, each enzyme peaking at its own T_{opt}. **Right:** folded fraction $f_N(T)$, each collapsing at its own T_m.

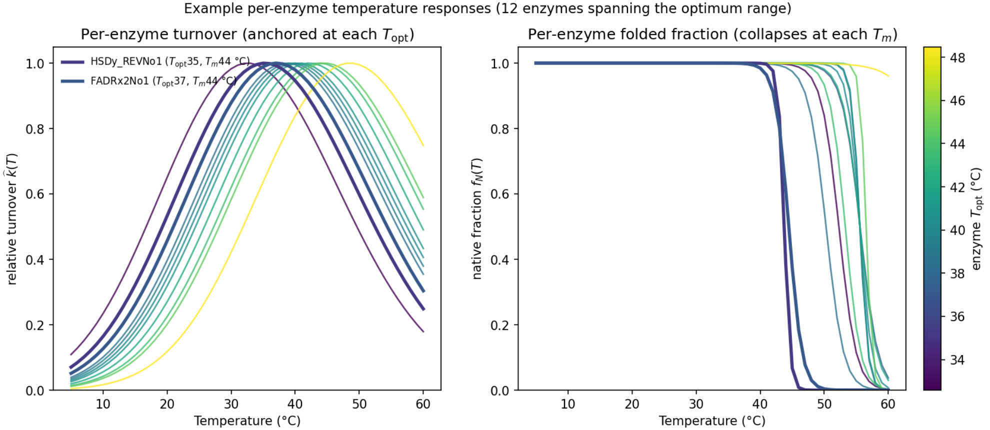

Figure 1: ~12 *E. coli* enzymes. `reports/ecoli_tpc`

The state of the art: Li et al. 2021

- Yeast `ecYeast7`; **764 enzymes, 2 292 parameters** (T_m, T_{opt}, $\Delta C_p^{\ddagger}$ each)
- T_m **measured** (266 of 764); T_{opt} **predicted** from sequence, carried as a wide prior
- Fitted to **growth and chemostat fluxes**
- **No gas exchange** — no O_2, no acetate
- Headline: ERG1 controls flux at 40 °C, confirmed in the wet lab[1]

What one of these curves looks like

Every descriptor a thermal biologist wants is read off **one predicted curve** — and each is a different question of the model.

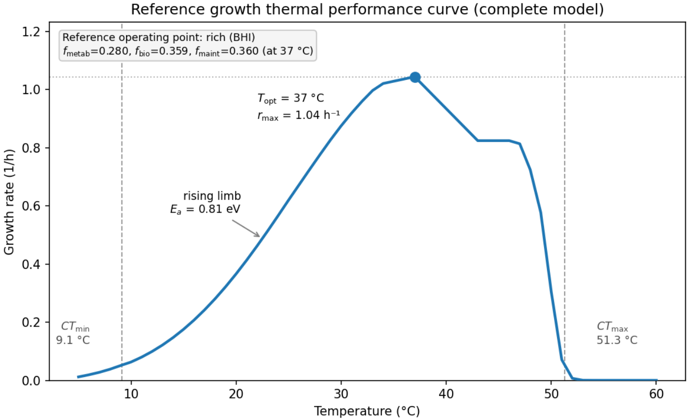

Figure 2: Our *E. coli* model at its reference point. `reports/ecoli_tpc`

The problem the field found: Pettersen & Almaas 2023

- *"the Bayesian calculation method … is* ***unstable and unable to estimate the posterior distribution****"* — it assumes one peak; the problem has many
- Of six signature reactions, the one that varies most between **equally-fit** solutions is **ferrocytochrome-c:oxygen oxidoreductase** — *"for some particles used extensively, in other cases not at all, still giving rise to approximately the same growth rates"*
- Their diagnosis and their ask:
  - *"the external measurements [are] insufficient to account for the inner workings of the … cell"*
  - *"measurements of proteomics and* ***fluxomics*** *will help narrow down the solution space"*[2]

The most under-determined reaction is the O_2-consuming step of respiration, and what they ask for is a measured flux. Hold that thought.

Where we started: the MRes baseline

- `iJO1366`, enzyme-constrained with GECKO 3 in MATLAB; **1 366 enzymes**
- **4 101 thermal parameters** — *more* per-enzyme freedom than Li et al.[1] — with T_m measured for 841 and T_{opt} from a database for 894
- Same thermal physics as now: two-state unfolding, transition-state k_{cat} with $\Delta C_p^{\ddagger}$, temperature-dependent maintenance
- **Not Bayesian**: repeated FBA scored by R^2 against **one** growth curve, the other 25 held out
- $R^2 = 0.94$; growth TPCs only[3]

Two things to be clear about: the GEM was later **replaced** (`iJO1366` → `eciML1515`), and the baseline already had per-enzyme parameters — what it had no way to do was **constrain** them.

What the model is now

- One shared core in Python; **seven organisms** on it (*E. coli* best-constrained)
- **16 global levers**, deliberately — an identifiable handful in place of 4 101 unconstrainable ones
- Everything is under a regression suite — **79 structural checks and 60 numerical ones**, re-run on every change, so a fix to one organism cannot silently move another
- The figure is **a priori**: nothing was fitted to that curve

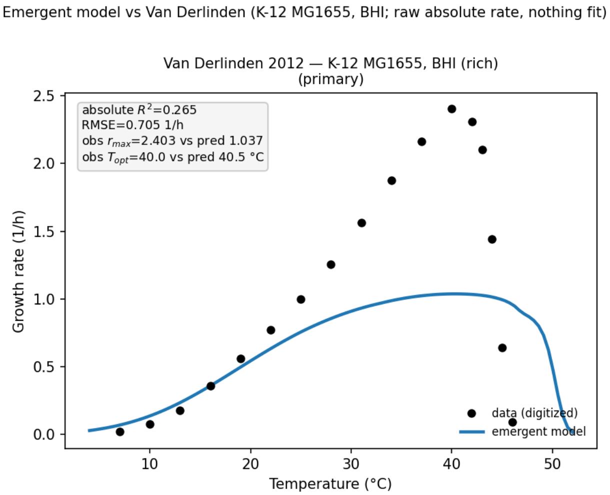

Figure 3: Van Derlinden et al.[5], K-12 MG1655 in BHI. `reports/ecoli_tpc`

The 16 levers — 1: the thermal envelope (6)

Each acts on **every** enzyme at once, on top of the per-enzyme data.

lever	what it does	what it is for
dT_{opt}	shifts every enzyme's T_{opt} by a constant	the database's optima are systematically off
`topt_scale`	stretches the *spread* of optima about 30 °C	how synchronised the enzymes are
ΔC_p `_scale`	scales the curvature of $k_{cat}(T)$	how broad each enzyme's response is
dT_m	shifts every melting temperature	*in vitro* $T_m \neq$ *in vivo* inactivation
`tm_scale`	stretches the *spread* of melting points	sharp collapse vs gradual shoulder
`kcat_scale`	scales the *level* of all turnover	the *in vitro* $\to$ *in vivo* k_{cat} gap[6]

The 16 levers — 2: budget, maintenance, medium, observation (10)

lever	what it does
σ	average *in vivo* enzyme saturation — how much of the protein is working
`kappa_scale`	effective translation efficiency in the biosynthesis cap
f_{metab}, f_{maint}	the proteome split between flux-carrying and housekeeping sectors
`ngam_scale`, `ngam_steepness`	size of the maintenance cost, and how fast it rises with T
`clearance_mult`	how fast the medium is actually drawn down (a *medium* lever)
`resp_scale`	mmol O_2 $\rightarrow$ measured units: an **observation** scale, not biology
`disc_resp`, `disc_growth`	model discrepancy — how much the model is allowed to be wrong

Which of the sixteen the data can actually move

Three kinds, and the difference matters — not two.

- **Eight the data move.** dT_{opt}, `topt_scale`, ΔC_p`_scale`, dT_m, `tm_scale`, `kcat_scale`, σ, `ngam_scale`
- **Two are FIXED BY MEASUREMENT** — f_{metab}, f_{maint}. They enter from the measured proteome, matched to medium *and* temperature, and carry **tight priors: measurement wiggle only**. The likelihood not moving them within that width is **the design working**, not a deficiency — growth and respiration are not asked to re-derive a proteomics measurement
- **Two are genuinely unidentified.** `kappa_scale` (effective translation demand, derived once at build time — nothing measures it independently) and `ngam_steepness` (maintenance steepness, *borrowed* from the MRes model). **These** are the ones a measurement would fix
- **Four are not model levers at all.** `clearance_mult` is a property of the experiment; `resp_scale`, `disc_resp`, `disc_growth` are the observation model

Development 1: proteome allocation, from measured proteomes

- The enzyme budget is split into three **sectors**[7,8]: **metabolic** (carries flux) · **biosynthesis** (ribosomes) · **maintenance** (chaperones, housekeeping)
- The split is **not fitted and not hand-set**. It is read from a study that *measured the E. coli proteome at a series of temperatures*, in separate LB / glucose / glycerol series[9]
- So the model is condition-specific for free: measured ribosome fraction at 37 °C is **0.36 on rich LB** but **0.19 on glucose-minimal** — a rich-medium run automatically gets a higher biosynthesis cap and grows faster
- The **maintenance** sector carries a measured **2.7-fold heat-stress chaperone rise**
- New relative to Li et al.[1] and to the MRes[3]: **neither has an allocation layer at all**

The measured sectors

The chaperone sector (red) ramps sharply above 37 °C — the measured heat-stress response entering as an allocation change, not a fitted parameter.

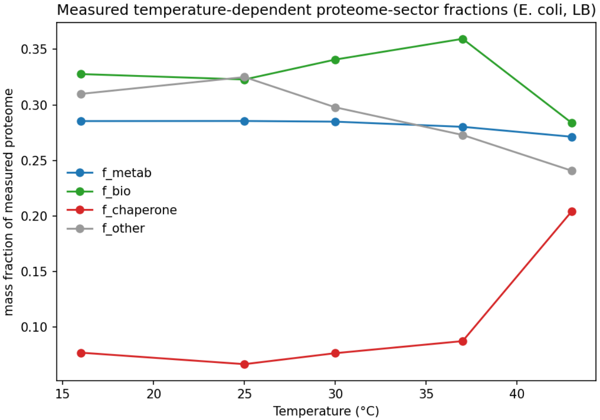

Figure 4: Wang et al.[9]; figure `reports/ecoli_tpc`

…and this is where we need more data

- **The measurement is thin where it matters most.** The glucose series was measured at **25, 30 and 37 °C only**; the LB series adds 16 and 43 °C
- Outside a series' range the allocation is **held at its endpoint** — so above 37 °C on glucose the sector split, and with it the translation cap, **stops varying with temperature**
- The consequence is visible in the curve: the glucose-minimal prediction is **bit-identical from 37 to 44 °C** (growth exactly 0.524757 h ^{1} at all eight grid points)
- **That shoulder is the proteomics table running out, not a property of the cell** — and it is why "T_{opt}" from such a run is the argmax of a tie

This is the concrete version of "we need more data": extend the measured glucose series above 37 °C. It is a proteomics experiment, not a modelling problem.

Which lever owns which feature of the curve

- The heatmap shows **12 rows, not 16**: `clearance_mult` is a property of the experiment, and `resp_scale`/`disc_resp`/`disc_growth` are the observation model — none of the four touches the predicted curve
- Read the rows in four groups:
  - **catalysis** $= \sigma +$ `kcat_scale` — how much working enzyme there is
  - **kinetic envelope** $= dT_{opt} +$ `topt_scale` $+ \Delta C_p$`_scale` — where the optima sit and how spread
  - **stability** $= dT_m +$ `tm_scale` — where unfolding takes over
  - **allocation / maintenance** $= f_{metab}$, f_{maint}, `kappa_scale`, `ngam_*`
- In lever terms: dT_{opt}/`topt_scale` move the **cold** limb and E_a; dT_m/`tm_scale` move the **hot** collapse; σ/`kcat_scale` move the **height** and nothing else
- **The result:** r_{max} is owned by catalysis (σ alone has elasticity ≈ 1); T_{opt} and CT_{max} by stability; E_a and niche width by the spread of optima
- **So a TPC descriptor is not a parameter.** Each one asks the model a different question — which is why fitting to peak height tells you nothing about the hot edge

Elasticity: the numbers

Fractional change in each descriptor per fractional change in each lever, at the operating point. **Deterministic — no intervals.** Bold is $|E| > 0.3$.

lever	T_{opt}	r_{max}	CT_{max}	niche width	E_a
dT_{opt}	0.00	−0.09	0.00	**−0.34**	**0.26**
`topt_scale`	0.00	−0.23	0.00	**−1.05**	**1.41**
ΔC_p`_scale`	0.00	−0.02	0.00	0.29	0.22
dT_m	0.13	0.09	0.13	0.18	−0.24
`tm_scale`	**−0.26**	−0.14	−0.16	−0.21	0.24
`kcat_scale`	0.00	**0.40**	0.00	0.07	**−0.37**
σ	0.00	**1.00**	0.00	−0.11	**0.36**
f_{maint}	0.00	**−0.56**	0.00	0.06	**−0.37**
`ngam_scale`	0.00	−0.00	0.00	−0.01	0.01
the three zero rows	0.00	0.00	0.00	0.00	0.00

…and the same numbers as a pattern

The three zero rows are zero for **different reasons**: `kappa_scale` and `ngam_steepness` unidentified, f_{metab} fixed by measurement.

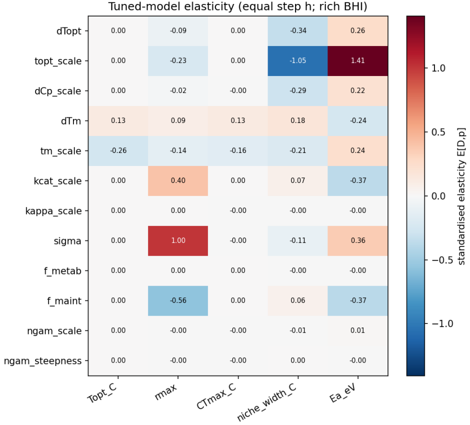

Figure 5: Elasticity heatmap, $h = 0.10$. `reports/ecoli_tpc`

Development 2: overflow metabolism — what it is

- Grow *E. coli* fast enough and it **excretes acetate while oxygen is still available** — it ferments and respires at once, wasting carbon. The bacterial analogue of the Crabtree effect
- Measured law: excretion is zero below a growth-rate threshold, then linear. **Threshold** ≈ 0.76 **h** ^{1}**, slope** ≈ 21 **mmol gDW** ^{1} **h** ^{1}[7]
- **Why any of this matters for a temperature model:** an enzyme-constrained TPC fixes the growth rate but *not* the internal flux distribution, so **O_2 and CO_2 exchange are under-determined** — the model can hit the right growth rate by respiring or by fermenting
- That is the *same* degeneracy Pettersen & Almaas found by FVA on the oxidase reaction[2], seen from the other side
- **Until it is pinned, an etcGEM cannot predict a gas flux at all**

The total-carbon cap: what it is

- **From the code:** $\sum_i n_{C,i}\, v_i \le c_{max}$ over **every open carbon uptake**, with $n_{C,i}$ the carbon count read from the model's own metabolite formula. Units **mmol C gDW ^{1} h ^{1}**; inorganic carbon is not charged to it
- **What it produces:** a finite carbon budget *and* a finite protein budget make the LP excrete acetate on its own. **Overflow emerges; it is not written in.** That is the difference between configuration D and configuration C
- **A second role, honestly stated:** an enzyme-constrained curve fixes growth but not the internal fluxes, so O and CO are under-determined. The cap is **one of three ways offered to pin them** — and it does **not** fully succeed: the O flux at the optimum still jumps between equally-optimal solutions, which is the cliff problem later in this deck

The total-carbon cap: why the value matters

- **At $c_{max} = 60$ acetate overflow is exactly zero on both media** — the cap suppresses the very mechanism it exists to create. On NLDM, T_{opt} also drops **39.0 → 32.5 °C** between 80 and 60
- **120 is the lowest value in his sweep at which overflow is non-zero on both** (23.5 on glucose, 3.9 on NLDM mmol gDW ^{1} h ^{1}) — which is why it was adopted
- **It is medium-dependent, and 60 / 120 / 450 is one quantity, not three guesses.** Applying the glucose/NLDM value of 120 to LB holds r_{max} at **1.1–1.8 h ^{1}** against a measured **2.945**, and growth R^2 falls to **0.64 / 0.05 / —0.10** across three configurations; at the LB-appropriate cap they are **0.90 / 0.83 / 0.88**
- **LB is settled at 450.** That was the resolution of a real disagreement, not a compromise

…and how it is made to emerge

Four ways to pin it — Parsa's configurations A–F, ported here and reproduced against his numbers.

	mechanism	imposed or emergent
A	cost the membrane carriers: transporters get the same thermal envelope as every other enzyme	emergent ceiling
B	a **total-carbon cap**, $\sum_i n_{C,i} v_i \leq c_{max}$	emergent
C	the measured acetate line written in as a constraint[7]	**imposed**
D	the carbon cap, **oxidase split left free** — the LP picks *bd* vs *bo3* per temperature	**emergent, T-dependent**

Why configuration D is the interesting one

- **Nothing is imposed on the acetate branch.** Overflow falls out of the protein budget plus a carbon budget — the model is not told that acetate appears above 0.76 h ^{1}, it produces it
- The **terminal-oxidase choice becomes a prediction**: *bd* and *bo3* have different melting temperatures, so which one the cell uses shifts with temperature on its own
- c_{max} is **not our number**: Parsa's own sweep (60 / 100 / 120 / 180 mmol C gDW ^{1} h ^{1}) concludes 100–120 is the sweet spot — *"a tight cap gives respiratory gas but a flat growth plateau; a loose cap gives a peaked TPC but fermentative gas"*. 120 adopted; 450 on LB
- Data behind the layer: the **acetate law**[7], the **membrane geometry** (Szenk et al., surface-to-volume 4.5 µm ^{1}), and Parsa's own **respirometry** for the fits
- **The advance over Li et al.[1] and the MRes[3]: both stop at growth.** This turns O_2, CO_2 and acetate into predictions that a measurement can check — and Parsa's respirometry is that measurement

Configuration D: the growth curve it predicts

The medium ladder is the measured sector split at work: r_{max} **1.19** (glucose) to **2.02** $\mathbf{h}^{\mathbf{1}}$ (BHI/LB), T_{opt} 38–40 °C. **Model output; no measurement plotted.**

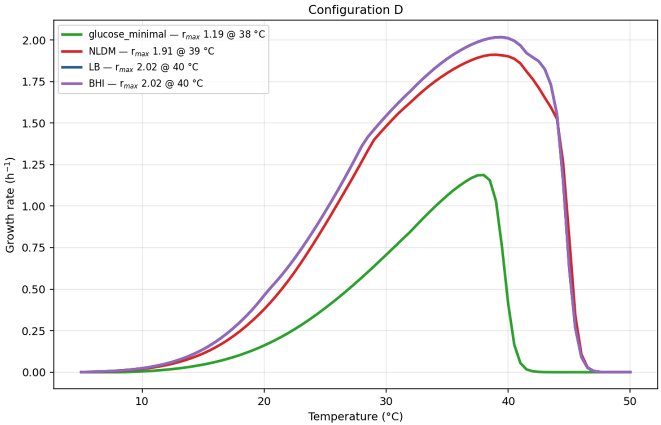

Figure 6: `reports/ecoli_gasflux`

…and the gas exchange it predicts

Right-hand panel: RQ ≈ 1 around the optimum — respiratory, not fermentative. **Model output; no measurement plotted; mechanism, not interval.**

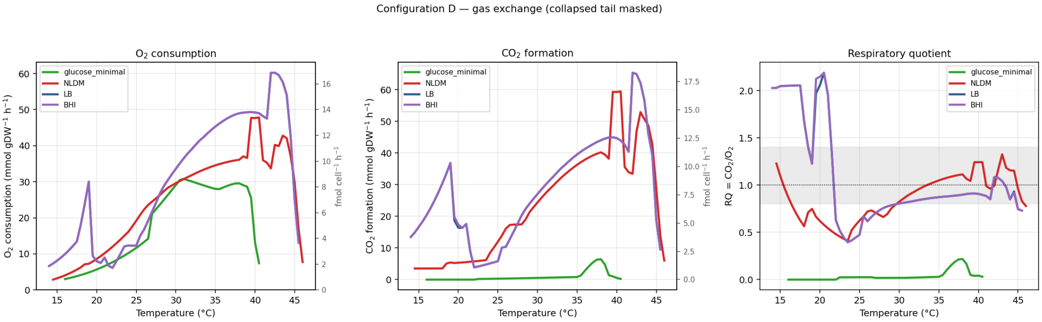

Figure 7: `reports/ecoli_gasflux`

The data nobody else has

- Parsa's respirometry: **O_2 flux and growth per cell, 15–50 °C, three media** (R2A, LB, M9), ∼60 replicate runs each
- This is exactly the **fluxomics on the O_2-consuming reaction** that Pettersen & Almaas[2] name as the missing data
- And it is a **measurement**, not a model output — which is what makes it able to break the degeneracy
- Ported and **checked against his own results**: all sixty of his configuration numbers and all ten of his respirometry fits reproduce here, the worst to within 0.009
- Look at the shape: **respiration peaks 3–5 °C above growth** (43–45 °C vs ∼40 °C) and is still climbing where growth has begun to fail

Measured growth and respiration, three media

Respiration peaks 3–5 °C above growth and is still climbing where growth has begun to fail. Faint points are replicate runs; bars s.e.m.

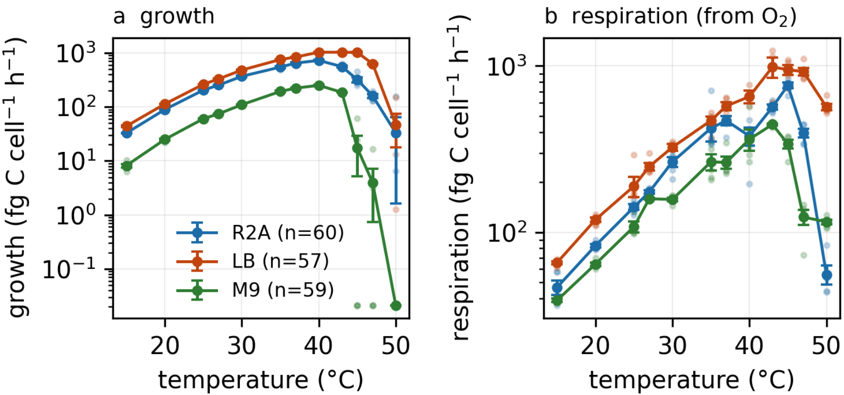

Figure 8: `strains/eciML1515/respirometry/`

How well does it fit? Growth

Lines are the model, points the measurement; two media, three configurations. **At Parsa's gated parameters — not a converged posterior, and no intervals.**

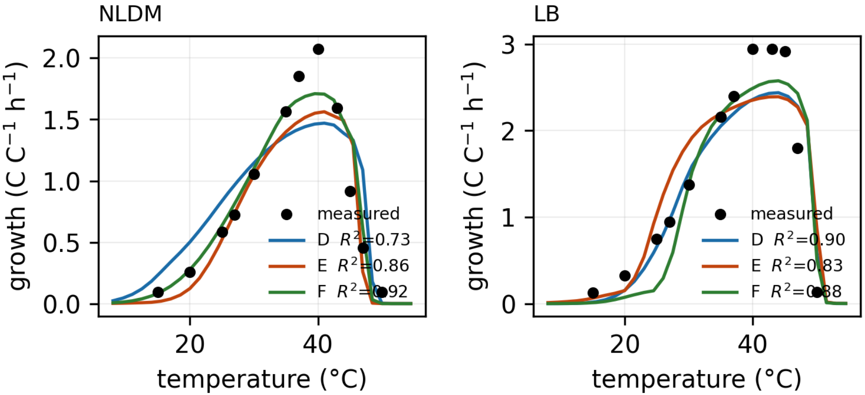

Figure 9: R^2 from `reports/P3_gate/gate_def_headline.csv`. `make_fit_figures.py`

How well does it fit? Oxygen

Gated parameters, no posterior. A **fitted** scale converts model to measurement: this tests **shape**, not level. **The jagged cold limb is the vertex-jumping from the cliff slides.**

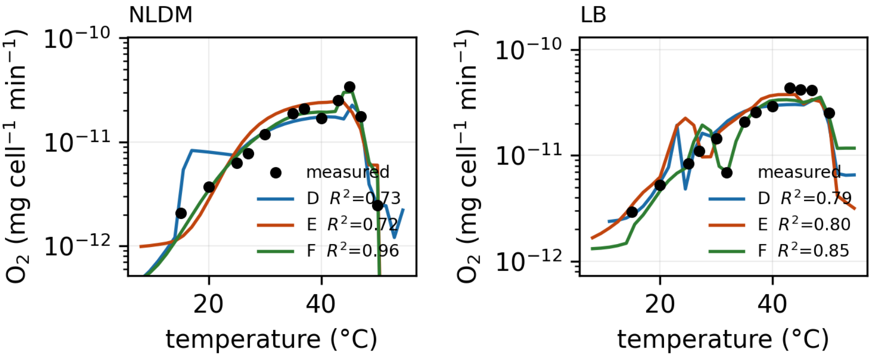

Figure 10: `make_fit_figures.py`

...and carbon-use efficiency, which we cannot yet overlay

- The model would need **biomass carbon over total carbon consumed**
- The second term is a sum over the **64 open carbon sources** of the NLDM recipe, which the prediction path does not return; the first needs a **biomass carbon fraction** that is not in the strain data
- Both would have to be invented, so **CUE stays measurement-only** — the measured curve is two slides back
- **This is the next thing to close**, and it is a small piece of plumbing rather than a new experiment

…and what it costs the cell

- **Carbon-use efficiency** — the share of assimilated carbon kept as biomass rather than respired
- Flat at ~**0.6** (R2A, LB) and ~**0.4** (M9) from 25 to 40 °C
- Then a **cliff**: 0.29 / 0.52 / 0.05 at 45 °C
- Medium sets the *level*; temperature sets the *cliff*
- **The model does not yet predict it.** CUE needs the carbon budget closed — biomass carbon against total carbon consumed — and that is the next thing the gas-exchange layer can be asked for

Measured carbon-use efficiency

Flat from 25 to 40 °C, then a cliff — and the medium sets the level, not the shape.

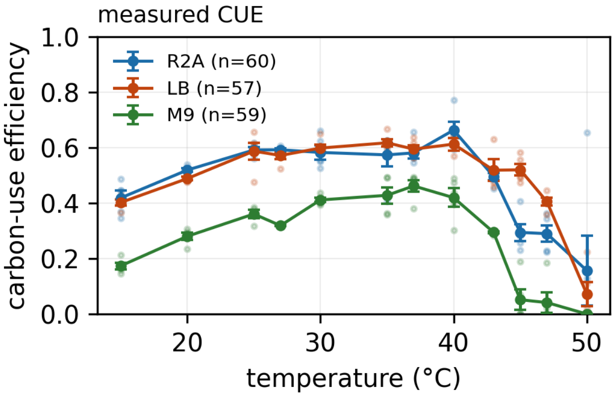

Figure 11: `strains/eciML1515/respirometry/`

The statistical problem, in plain terms

What everyone is trying to do: the per-enzyme parameters are uncertain, so treat them as unknowns and ask the data which values are consistent with it — a *distribution* over parameters, not one best guess.

- **Li et al.** used **SMC-ABC**: draw parameter sets from the prior, simulate, keep the ones whose predictions land close to the data, narrow the tolerance, repeat. Reported a posterior over 2 292 parameters[1]
- **Pettersen & Almaas** re-ran it **from different random seeds and got different answers**. The method assumes the good parameter sets form a single hill; in this problem there are many. They swapped in an **evolutionary algorithm** to collect a *diversity* of good solutions instead — then showed those solutions disagree wildly on internal fluxes[2]
- **What we do differently: 16 parameters, not 2 292** — small enough for a proper MCMC sampler (an ensemble sampler) rather than an approximate one — and a likelihood that includes the **measured respiration**, not growth alone

…and the three things that went wrong

1. **The chains are too short.** Every chain in this family — ours and Parsa's — ran about **9 autocorrelation times** where the standard is ≥ 40. So **no interval on any parameter is available**, and none appears in this deck. Fixing it is arithmetic: 4–5$\times$ the chain length
2. **The surface itself is not smooth.** We stopped blaming the sampler and looked at the likelihood: 22 line scans, 902 model builds, **12 of 22 lines jump by >20 % of their range on a small step**. Not the solver — unchanged at 10^{-9} tolerances and on a different algorithm
3. **The cause is the respiration term, and it is the degeneracy again.** At cold temperatures the O_2 flux at the growth optimum **jumps between equally-optimal solutions** — so the prediction moves discontinuously while the growth rate does not. *Exactly* what Pettersen & Almaas saw by FVA[2], here as cliffs in a likelihood

- **Three fixes, each for a named mechanism:** a **tie-break** to choose one point on the degenerate face; a **variance floor** so the likelihood is not shocked by a jump the model cannot avoid; a **continuous support** clamp at the growth threshold. Cliffs cut ~10$\times$. **Still not smooth, and we say so**

The surface, scanned

One panel per parameter line — read the pattern, not the axes: parabolas broken by cliffs, and two panels flat all the way across.

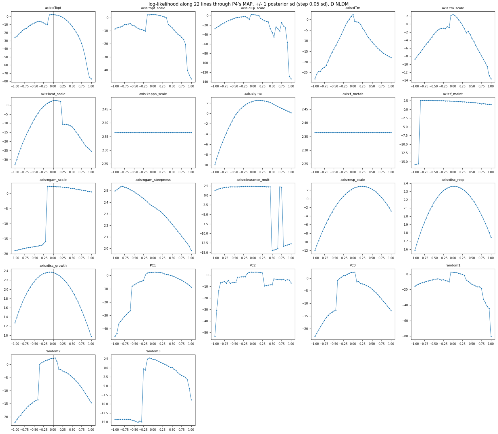

Figure 12: 22 line scans, 902 model builds. `reports/P9_surface`

How many answers are there? — and two retractions

- We then mapped the **basins**: run the optimiser from many starting points, cluster where it lands, and ask how many distinct good solutions there are
- **Result: one live basin and one dead one**, of twelve converged endpoints. **Not** the many-modes picture Pettersen & Almaas[2] found in yeast — but also **not yet a posterior**
- **Retraction 1 (ours).** We had concluded the model *cannot* fit these data while honouring the measured melting temperatures. **It can**: with both melting-temperature levers pinned to the measurement the fit converges and the organism grows at 1.66 h ^{1} — 7.6 log-likelihood units worse, and pressed against two prior bounds
- **Retraction 2 (ours).** The test we used to claim the optimiser reaches the top of the basin was **partly an artefact**: five of twelve endpoints park a temperature on an internal threshold that acts as an attractor

Where the posterior stands

- **Two runs are in flight.** The stopping rule was fixed **before** they started: chain length > 40 autocorrelation times **and** effective sample size ≥ 200
- **Nothing is quoted from them here** — that is the whole point of pre-registering the criterion
- What we will be able to say when they land: an interval on each of the sixteen levers, conditional on the mode, and a posterior-predictive band on the TPC and on the gas fluxes
- What we will **still** not know: `kappa_scale` and `ngam_steepness` — those need a measurement, not a longer chain

The honest position today: mechanisms that reproduce, a likelihood we understand, and no intervals.

Auditing the published model

- We built an audit for a defect class we had found in our **own** models — a free ion path that lets ATP synthase run without the respiratory chain paying for it — and ran it on Li et al.'s deposited model[1]
- **The defect is absent.** Their respiratory chain supplies **93 %** of ATP synthase's protons; all twelve uncosted proton-movers run the dissipative way; **no free path exists at any length**
- Then we repeated one of our own structural findings at **their published posterior** — 100 models, their code, their parameters: changing which constraint binds widens the top of the TPC from **1.5 to 7.3 °C in 93 %** of their models
- **But CT_{max} also moves (4.6 °C).** In *our* model it did not. **The asymmetry is not the clean thing we thought it was**
- Register: an audit applied to the reference implementation — not a critique of a paper

The asymmetry at Li et al.'s own posterior

Left: the top of the curve widens from ~1.5 to ~7.3 °C. **Right:** T_{opt} moves further than CT_{max} in 92 % of their models — but CT_{max} is **not** immune.

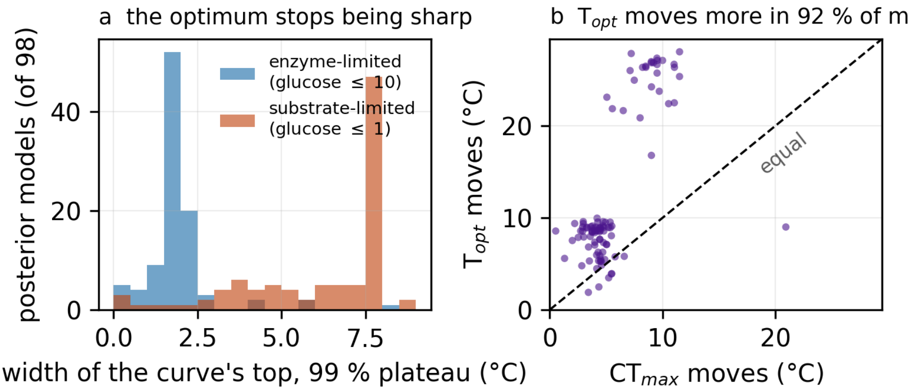

Figure 13: `reports/Y2_regime_posterior`

The awkward one: our model needs a −4 K stability shift

- Every fit of the *E. coli* model wants melting temperatures **3–4.5 K below a measured meltome**[10]
- Two readings, and we screened them: **biology** (*in vitro* T_m is not *in vivo* inactivation) or **our parameterisation** (16 global levers where Li et al.[1] had 2 292)
- **The shift is not uniquely required** — an equally good, living fit exists at zero shift, for 0.08 log-likelihood units
- **But catalysis is not what pays for it.** The compensator is `tm_scale`, the *spread* of melting temperatures. Both solutions put the least-stable few per cent of enzymes **4–6 °C below measurement**
- So the **number** is a parameterisation artefact; the **contradiction with the meltome's low tail** is not. Per-enzyme $k_{cat}(T)$ would not fix it — and the predictor we would have used has no usable optimum for 97 % of these enzymes

Settled · in flight · needs measurement

- **Settled.** One shared codebase with seven organism models on it, under the regression suite. The mechanisms: measured proteome sectors, overflow emerging from a carbon cap, the ETC area budget. The audits — including the reference implementation passing ours
- **Settled, and uncomfortable.** No interval from the older *E. coli* fits is available. The likelihood surface is structurally rough. The meltome's low tail is contradicted by every fit
- **In flight.** The posterior, on a pre-registered convergence criterion
- **Needs measurement — not more modelling.** Proteome allocation **above 37 °C on glucose** — the measurement exists, its *range* does not, and that is where CT_{max} is set; maintenance vs temperature from a low-dilution chemostat (`ngam_*` are borrowed); *in vivo* stability for the enzymes that set the thermal limit
- **Never yet tested.** Nothing in this framework has been asked to predict data it was not fitted to. Cooper et al.'s evolved *E. coli* lines[11] are the obvious holdout

Three questions for the group

1. **Is a few-kelvin gap between an *in vitro* meltome and *in vivo* functional inactivation expected?** If it is, the model is asking the meltome the wrong question and we should say so. If it is not, the model is wrong somewhere we have not looked
2. **Respiration peaks 3–5 °C above growth, and carbon-use efficiency falls off a cliff at 43–45 °C.** Is that a general property of bacterial thermal performance, or a feature of these three media?
3. **What would you accept as a real test?** Our candidate is predicting how a TPC *shifts* after 20 000 generations of thermal selection[11]. Is there a better one?

References I

1. Li, G. *et al.* Bayesian genome scale modelling identifies thermal determinants of yeast metabolism. *Nature Communications* **12**, 190 (2021).

2. Pettersen, J. P. & Almaas, E. Parameter inference for enzyme and temperature constrained genome-scale models. *Scientific Reports* **13**, 6079 (2023).

3. Madkaikar, A. Predicting the thermal niche of a ubiquitous bacterium using whole genome sequence. (Imperial College London, 2023).

4. Hobbs, J. K. *et al.* Change in heat capacity for enzyme catalysis determines temperature dependence of enzyme catalyzed rates. *ACS Chemical Biology* **8**, 2388–2393 (2013).

5. Van Derlinden, E. & Van Impe, J. F. Modeling growth rates as a function of temperature: Model performance evaluation with focus on the suboptimal temperature range. *International Journal of Food Microbiology* **158**, 73–78 (2012).

References II

6. Davidi, D. *et al.* Global characterization of in vivo enzyme catalytic rates and their correspondence to in vitro kcat measurements. *Proceedings of the National Academy of Sciences* **113**, 3401–3406 (2016).

7. Basan, M. *et al.* Overflow metabolism in Escherichia coli results from efficient proteome allocation. *Nature* **528**, 99–104 (2015).

8. Scott, M., Gunderson, C. W., Mateescu, E. M., Zhang, Z. & Hwa, T. Interdependence of cell growth and gene expression: Origins and consequences. *Science* **330**, 1099–1102 (2010).

9. Wang, D. *et al.* A proteome optimal allocation model for elucidating effects of temperature on bacterial growth. *npj Systems Biology and Applications* https://doi.org/10.1038/s41540-026-00693-4 (2026) doi:10.1038/s41540-026-00693-4.

10. Leuenberger, P. *et al.* Cell-wide analysis of protein thermal unfolding reveals determinants of thermostability. *Science* **355**, eaai7825 (2017).

References III

11. Cooper, V. S., Bennett, A. F. & Lenski, R. E. Evolution of thermal dependence of growth rate of Escherichia coli populations during 20,000 generations in a constant environment. *Evolution* **55**, 889–896 (2001).